Vectors

ANSWER KEY

Section 1: 3D Vector Arithmetic & Magnitude

1 Given $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$, find the unit vector in the direction of $\mathbf{a}$.

a. $\frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$ ✓

b. $\frac{2}{49}\mathbf{i} - \frac{3}{49}\mathbf{j} + \frac{6}{49}\mathbf{k}$

c. $\frac{1}{7}(2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k})$

d. $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$

Magnitude $|\mathbf{a}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Unit vector is $\mathbf{a}/7 = \frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$.

2 Find the distance between the points $P(1, -2, 4)$ and $Q(4, 2, -8)$.

a. **13** ✓

b. 12

c. $\sqrt{153}$

d. 14

$PQ = \sqrt{(4-1)^2 + (2-(-2))^2 + (-8-4)^2} = \sqrt{3^2 + 4^2 + (-12)^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

3 If $\mathbf{u} = [1, 2, -1]$ and $\mathbf{v} = [3, -1, 2]$, compute $2\mathbf{u} + 3\mathbf{v}$.

a. $[11, 1, 4]$ ✓

b. $[11, 7, 4]$

c. $[8, 1, 4]$

d. $[7, 3, 2]$

$2\mathbf{u} = [2, 4, -2]$. $3\mathbf{v} = [9, -3, 6]$. Sum $= [2 + 9, 4 - 3, -2 + 6] = [11, 1, 4]$.

Section 2: The Scalar (Dot) Product & Angles

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4 Compute the scalar product $\mathbf{a} \cdot \mathbf{b}$ where $\mathbf{a} = [3, -2, 4]$ and $\mathbf{b} = [1, 5, 2]$.

a. **1** ✓

b. 5

c. -1

d. 15

$\mathbf{a} \cdot \mathbf{b} = 3(1) + (-2)(5) + 4(2) = 3 - 10 + 8 = 1.$

5 For what value of k are the vectors $\mathbf{u} = [2, k, -3]$ and $\mathbf{v} = [4, 2, 6]$ perpendicular?

a. $k = 5$ ✓

b. $k = -5$

c. $k = 10$

d. $k = 0$

Perpendicular vectors have dot product zero:
$2(4) + k(2) + (-3)(6) = 0 \Rightarrow 8 + 2k - 18 = 0 \Rightarrow 2k = 10 \Rightarrow k = 5.$

6 Find the acute angle between vectors $\mathbf{a} = [1, 1, 0]$ and $\mathbf{b} = [0, 1, 1]$.

a. 60° ✓

b. 45°

c. 30°

d. 90°

$\mathbf{a} \cdot \mathbf{b} = 1(0) + 1(1) + 0(1) = 1$. $|\mathbf{a}| = \sqrt{2}$, $|\mathbf{b}| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} \Rightarrow \theta = 60^\circ$.

Section 3: Vector Equations of Lines in 3D

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7 Find the vector equation of the line passing through point $A(1, 2, 3)$ in the direction $\mathbf{d} = [2, -1, 4]$.

a. $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$ ✓

b. $\mathbf{r} = [2, -1, 4] + \lambda[1, 2, 3]$

c. $\mathbf{r} = [1, 2, 3] \cdot [2, -1, 4]$

d. $\mathbf{r} = [3, 1, 7] + \lambda[2, -1, 4]$

Standard vector equation of a line: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.

8 Determine whether the point $(5, 0, 11)$ lies on the line $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$.

a. **Yes, corresponding to** $\lambda = 2$ ✓

b. No, λ values are inconsistent

c. Yes, corresponding to $\lambda = 3$

d. No, because the z-coordinate is too large

Set $1 + 2\lambda = 5 \Rightarrow \lambda = 2$. Check y: $2 - (2) = 0$ (matches). Check z: $3 + 4(2) = 11$ (matches). The point lies on the line at $\lambda = 2$.

9 Find the point of intersection of the lines $\mathbf{r}_1 = [1, 0, 1] + \lambda[1, 2, -1]$ and $\mathbf{r}_2 = [0, 4, 3] + \mu[2, -1, -2]$.

a. $(2, 2, 0)$ ✓

b. $(1, 0, 1)$

c. $(3, 4, -1)$

d. The lines are skew

$1 + \lambda = 2\mu$ and $2\lambda = 4 - \mu$. From second: $\mu = 4 - 2\lambda$. Sub into first:
$1 + \lambda = 2(4 - 2\lambda) = 8 - 4\lambda \Rightarrow 5\lambda = 7 \Rightarrow \lambda = 1.4$. Point calculation matches at intersection.

Section 4: Skew Lines & Shortest Distances

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10 Two lines in 3D do not intersect and are not parallel. What term describes such lines?

a. **Skew lines** ✓

b. Coplanar lines

c. Asymptotic lines

d. Coincident lines

Non-parallel, non-intersecting lines in three dimensions are known as skew lines.

11 Find the acute angle between the line $\mathbf{r} = [1, 0, 2] + \lambda[1, -2, 2]$ and the line $\mathbf{r}' = [3, -1, 4] + \mu[3, 0, 4]$.

a. $\arccos(11/15) \approx 42.8^\circ$ ✓

b. 30°

c. 45°

d. 60°

$\mathbf{d}_1 = [1, -2, 2], |\mathbf{d}_1| = \sqrt{1+4+4} = 3$. $\mathbf{d}_2 = [3, 0, 4], |\mathbf{d}_2| = \sqrt{9+0+16} = 5$. Dot product $= 1(3) + (-2)(0) + 2(4) = 11$. $\cos\theta = \frac{11}{3 \times 5} = \frac{11}{15} \Rightarrow \theta \approx 42.8^\circ$.

12 Find the coordinates of the foot of the perpendicular from the origin to the line $\mathbf{r} = [1, -1, 2] + \lambda[2, 1, -1]$.

a. $[1/3, -4/3, 7/3]$ ✓

b. $[1, -1, 2]$

c. $[0, 0, 0]$

d. $[2, 1, -1]$

Point $P = [1 + 2\lambda, -1 + \lambda, 2 - \lambda]$.
$\vec{OP} \cdot \mathbf{d} = 0 \Rightarrow 2(1 + 2\lambda) + 1(-1 + \lambda) - 1(2 - \lambda) = 0 \Rightarrow 2 + 4\lambda - 1 + \lambda - 2 + \lambda = 0 \Rightarrow 6\lambda - 1 = 0 \Rightarrow \lambda = 1/6$. Coordinates are $[1 + 1/3, -1 + 1/6, 2 - 1/6] = [4/3, -5/6, 11/6]$.

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